

Research Profile

I build robots that learn through physical interaction, and the real-world systems that make that learning work.

Education

University of Maryland, College Park — Ph.D. in Computer Science Aug. 2022–present
Perception and Robotics Group; advised by Yiannis Aloimonos and Cornelia Fermüller.

Nanjing Institute of Technology — B.Eng. in Robotics Sep. 2018–Jul. 2022

Research & Industry Experience

Amazon Frontier AI & Robotics Team — Research Intern Jan. 2026–present

- Developing learning systems for force-aware dexterous manipulation from multimodal human interaction.
- Exploring physics-grounded, interaction-aware video and world models for robot learning.
- Worked on large-scale contact-aware data labeling for robot-learning datasets.

Honda Research Institute USA — Research Intern May–Aug. 2025
Developed robust motion prediction using a trainable mixture of model-based and learned experts.

Carnegie Mellon University — Research collaboration May 2023–2025
Interactive navigation and field autonomy; collaborated with Ji Zhang.

Selected Research & Systems

Dexterity from Supervised Play — learn dexterity from minutes of goal-agnostic human play
A framework in which the robot learns a goal-agnostic play policy from minutes of human play, then deploys it to autonomously collect hours of contact-rich interaction data for training an action-conditioned world model. At inference, model-predictive control selects among play-policy action proposals to reach a requested goal, without task-specific demonstrations.

ForceBand — bring force into human data with EMG, for forceful manipulation
ForceBand is a low-cost, wrist-worn sEMG band that predicts fingertip forces from forearm muscle activity, enabling force-enriched human demonstrations for learning forceful robot manipulation—without any force sensors on the hand.

FEEL — force-enhanced egocentric dataset
A large-scale dataset pairing egocentric video with calibrated glove-based force measurements, comprising approximately three million synchronized frames of natural manipulation.

HumanEgo — learning from minutes of human video
A robot-data-free, hardware-agnostic, and data-efficient pipeline that learns robot manipulation policies from minutes of raw human egocentric videos, powered by a flow matching policy with dense auxiliary objectives.

Interactive-FAR & Air-FAR — deployable field autonomy
Built navigation systems for ground and aerial robots operating in complex unknown environments. Interactive-FAR reasons about movable obstacles online; Air-FAR extends fast, adaptable routing to large-scale aerial navigation.

Microsaccade-Inspired Event Camera — sensing hardware and algorithms

Co-designed an actuated event-camera mechanism and computational pipeline that reveals static scene structure while preserving compatibility with established event-based perception methods.

Selected Publications

- **Botao He**, Yiming Dou, Yiannis Aloimonos, Jitendra Malik, Akash Sharma, Ruoshi Liu. *Dexterity from Supervised Play*. arXiv coming soon.
- **Botao He**, Zhi Wang, Linna Kuang, et al. *ForceBand: Learning Forceful Manipulation with sEMG*. CoRL 2026. [Project](#)
- Eadom Dessalene, **Botao He**, Michael Maynard, et al. *FEEL (Force-Enhanced Egocentric Learning): A Dataset for Physical Action Understanding*. ECCV 2026, **Long Oral (25/10,473 submissions)**. [Project](#) · [Paper](#)
- Zhi Wang, **Botao He**, Kelin Yu, et al. *HumanEgo: Zero-Shot Robot Learning from Minutes of Human Egocentric Videos*. IROS 2026 WORLDS Workshop **Best Paper**. [Project](#)
- Yu Chen, **Botao He**, Yuemin Mao, et al. *Adversarial Game-Theoretic Algorithm for Dexterous Grasp Synthesis*. Preprint, 2025.
- **Botao He***, Amir Hossein Shahidzadeh*, Yu Chen*, et al. *NavMoE: Hybrid Model- and Learning-based Traversability Estimation for Local Navigation via Mixture of Experts*. ICRA, 2025.
- **Botao He**, Guofei Chen, Cornelia Fermüller, Yiannis Aloimonos, Ji Zhang. *Air-FAR: Fast and Adaptable Routing for Aerial Navigation in Large-scale Complex Unknown Environments*. ICRA, 2025.
- **Botao He***, Guofei Chen*, Wenshan Wang, et al. *Interactive-FAR: Interactive, Fast and Adaptable Routing for Navigation among Movable Obstacles in Complex Unknown Environments*. IROS, 2024.
- **Botao He**, Ze Wang, Yuan Zhou, et al. *Microsaccade-Inspired Event Camera for Robotics*. *Science Robotics*, 2024.
- Qianhao Wang*, **Botao He***, Zhiren Xun, Chao Xu, Fei Gao. *GPA-Teleoperation: Gaze Enhanced Perception-aware Safe Assistive Aerial Teleoperation*. IEEE Robotics and Automation Letters 7(2), 2022; presented at ICRA 2022.
- **Botao He***, Haojia Li*, Siyuan Wu, et al. *FAST-Dynamic-Vision: Detection and Tracking Dynamic Objects with Event and Depth Sensing*. IROS, 2021.

* Equal contribution. Full publication record: [Google Scholar](#).

Technical Skills

Learning	Imitation learning, world models, diffusion policies, multimodal learning, online adaptation
Robotics	Dexterous manipulation, motion planning, navigation, active perception, ROS, Gazebo, AirSim, Unity
Programming	Python, C/C++, MATLAB, Git, Linux
Hardware	Custom sensing, embedded systems, STM32/Arduino, SolidWorks, machining, circuit design

Service & Recognition

Reviewing	Science Robotics; IEEE TPAMI; IEEE RA-L; IEEE T-ITS; ICRA; IROS; Frontiers in Robotics and AI
Recognition	NeuroPAC Fellowship (2023, 2024); University of Maryland Dean's Fellowship (2022–2023)